Sec 2.6

Rational functions

Def

A rational function is a function of the form

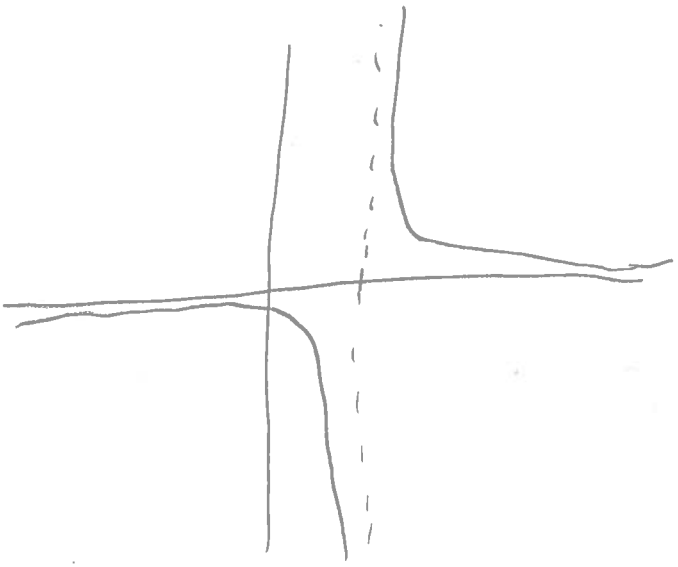
$$f(x) = \frac{N(x)}{D(x)}$$ where $N(x)$, $D(x)$ are Polynomials. Here we need to exclude the x-values where $D(x) = 0$

Definitions of vertical and Horizontal assymptotes.

① The line $x = a$ is a vertical assymptote of f when $f(x) \to \infty$ or $f(x) \to -\infty$ as $x \to a$

② The line $y = b$ is a Horizontal assymptote of f when $f(x) \to b$ as $x \to \infty$ or $x \to -\infty$

$f(x) = \frac{1}{x-1}$



$f(x) = \frac{1}{x-1}$ has a V.A. at $x = 1$ and Horizontal assymptotes at $x = 0$

Rules for vertical and Horizontal Assymptotes.

Let f be a rational function.

$$f(x) = \frac{N(x)}{D(x)} = \frac{a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0}{b_n x^n + b_{n-1}x^{n-1} + \cdots + b_1 x + b_0}$$

Where $N(x)$, $D(x)$ have no common factors,

Then

① the graph of $f(x)$ has vertical assymptotes (V.A.) at the Zeros of $D(x)$

② The graph of f has one or no Horizontal Assymptotes (H.A.) determined by comparing the degrees of $N(x)$, $D(x)$

Let the degree of $N(x) = n$ and the degree of $D(x) = m$ Then.

- when $n > m$ $f(x) = \frac{N(x)}{D(x)}$ has no Horizontal assymptotes
- when $n = m$ $f(x) = \frac{N(x)}{D(x)}$ has a horizontal assymptote at $y = \frac{a_n}{b_n}$ (ratio of leading coefficents)
- when $n < m$ $f(x)$ has a horizontal assymptote at $y = 0$

To get a sense for using these rules Let's do some Home work examples.

HW Example Line # 52 HW # 2

Find the H.A, V.A. x intercepts, y intercepts for

$$f(x) = \frac{x^2-16}{x^2-4}$$

The vertical assymptotes are where the bottom is zero

$$x^2-4=0 \Rightarrow x^2=4 \Rightarrow x=\pm\sqrt{4} \Rightarrow x=\pm 2$$

So we have

V.A.'s at $x=2$, $x=-2$

Here the degree of the top is the same as the degree of the bottom so we have a H.A. at $y=\frac{a_n}{b_n}=\frac{1}{1}$ The ratio of the leading coefficients.

HA: $y=1$

To find the x intercepts we want to know where the top is zero

$$x^2-16=0 \Rightarrow x^2=16 \Rightarrow x=\pm\sqrt{16} \Rightarrow x=\pm 4$$

So we have **x intercepts at $x=4$, $x=-4$**

The y intercept is where $x=0$ so we plug in 0 for x

$$f(0)=\frac{0^2-16}{0^2-4}=\frac{-16}{-4}=$$ **4 y int**

Before we sketch we want to know if the graph crosses the horizontal asymptote. We can set $f(x) = 1$ and solve:

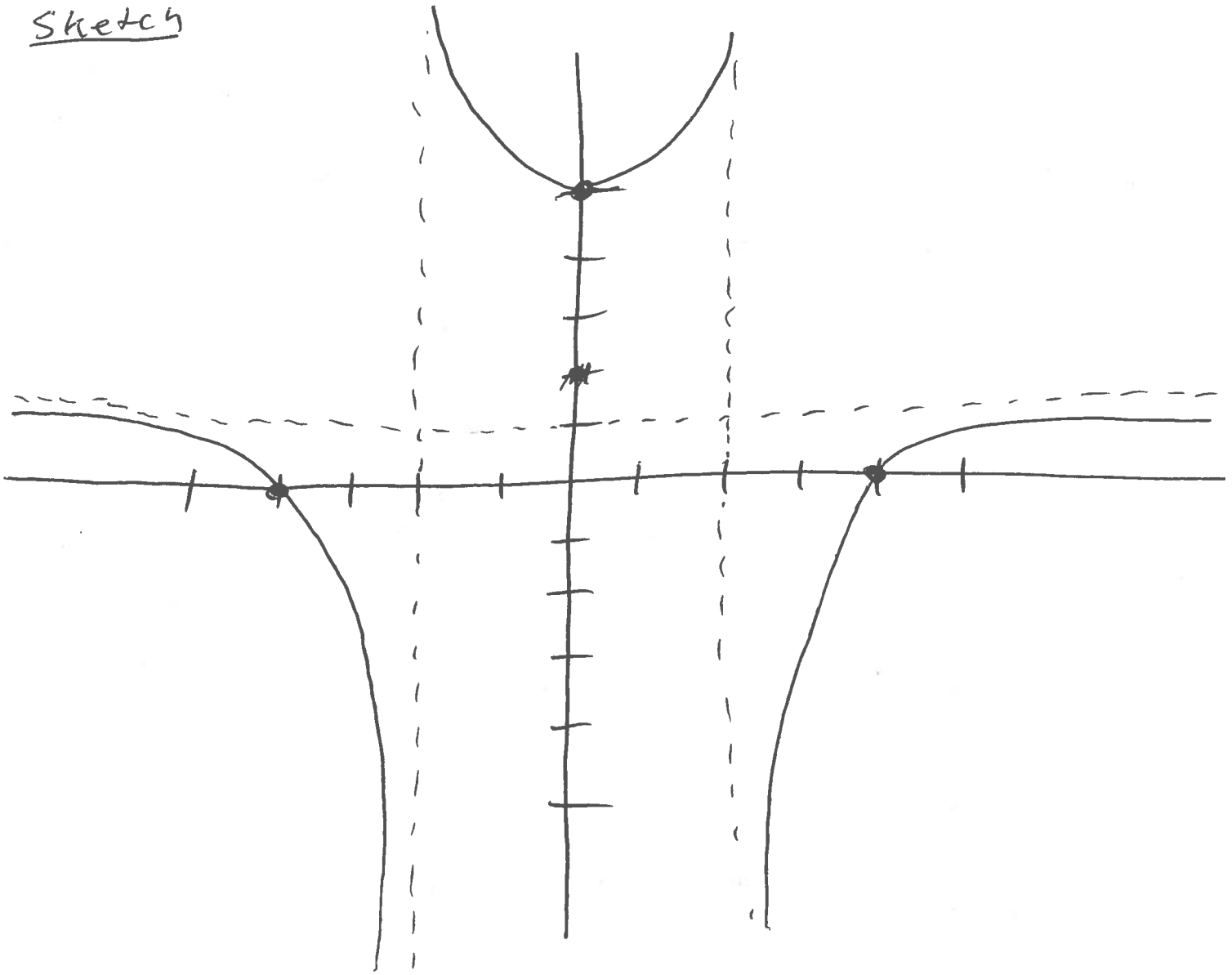
$$(x^2-4)\,1 = \frac{x^2-16}{x^2-4}(x^2-4)$$

multiply both sides by x^2-4

$$x^2-4 = x^2-16$$
$$-x^2 \qquad -x^2$$
$$-4 = -16 \leftarrow$$

This can't be true so we never crosses the HA.

Sketch



Like # 53 HW 2

$$f(x) = \frac{x^2-5x-36}{x^2-x}$$

V.A.'s where the bottom = 0

$$x^2-x = x(x-1) \Rightarrow x=0 \quad x=1$$

VA's at $x=0$ $x=1$

HA's The degree of the top is the same as the degree of the bottom so we have a Horizontal assymptote at $y = \frac{a_n}{b_n} = \frac{1}{1}$

at $y=1$

X int's where top = 0

$$x^2-5x-36 = 0$$

factor

$$(x-9)(x+4) = 0$$

$$\Rightarrow \boxed{x=9 \quad x=-4 \quad x\text{ int}}$$

Y ints plug in $x=0$

$$f(0) = \frac{0^2-5(0)-36}{0-0}$$

← No y int since 0 on bottom also V.A. at $x=0$

we also want to know if $f(x)$ crosses it's HA at $y=1$

$$1 = \frac{x^2-5x-36}{x^2-x}$$

$$x^2-x = x^2-5x-36$$

$$-x = -5x-36$$

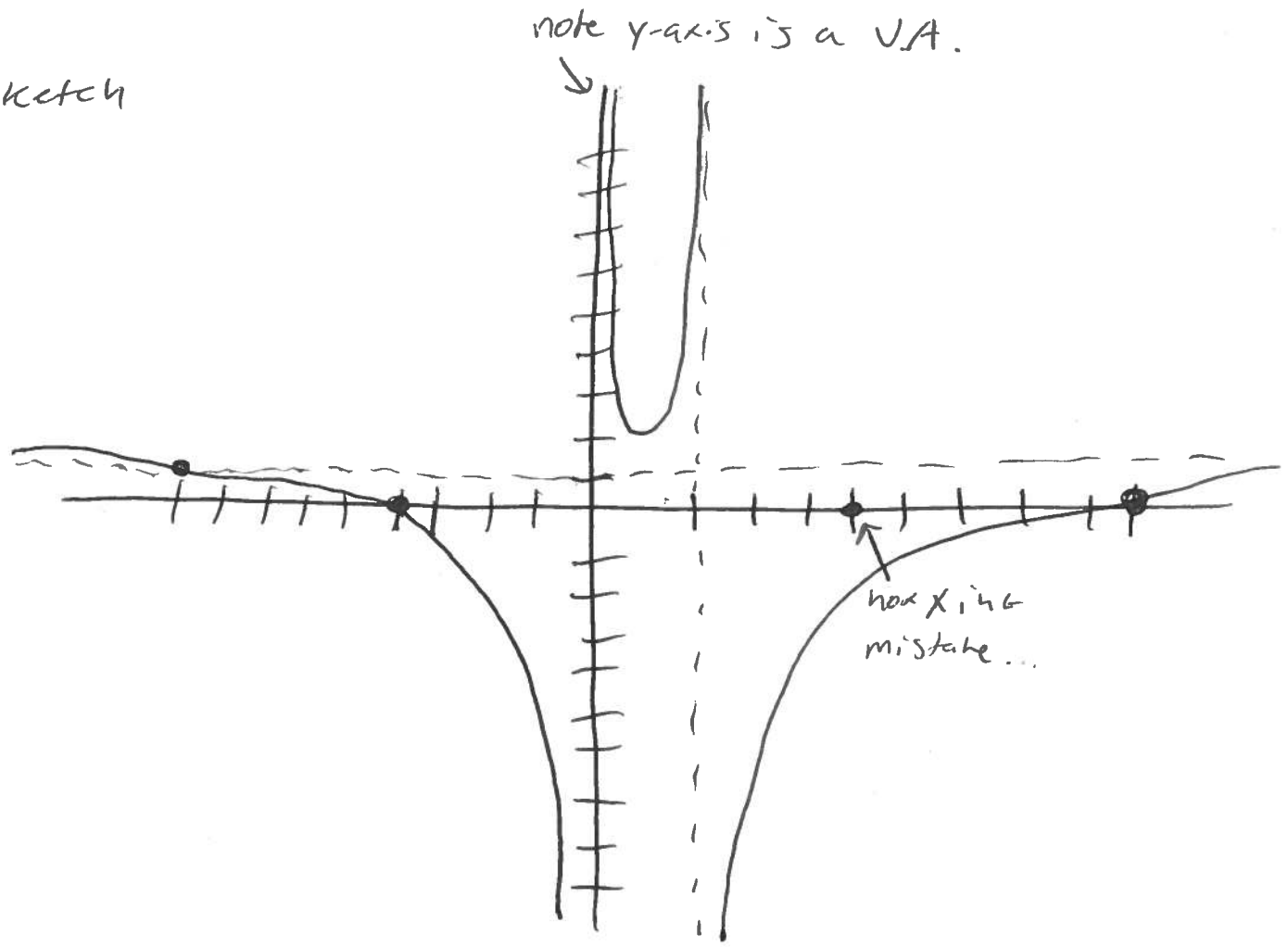
$$4x = -36$$

$$x = \frac{-36}{4}$$

$x=-9$ ← So at $x=-9$ we cross the HA.

9

Sketch



Like #55 Hw 2

$$f(x) = \frac{4x+8}{x^2+x-12}$$

V.A. where Bottom is zero

$$x^2+x-12=0$$

$$(x-3)(x+4)=0$$

$$\Rightarrow \boxed{x=3 \quad x=-4 \quad VA's}$$

H.A. Here the degree of the bottom is Bigger than that of the top so we have a Horizontal assymptote at $\boxed{y=0}$ the x-ax's

x int

where $f(x)=0$ same as where the top is zero

$$4x+8=0 \Rightarrow 4x=-8 \Rightarrow x=-8/4$$

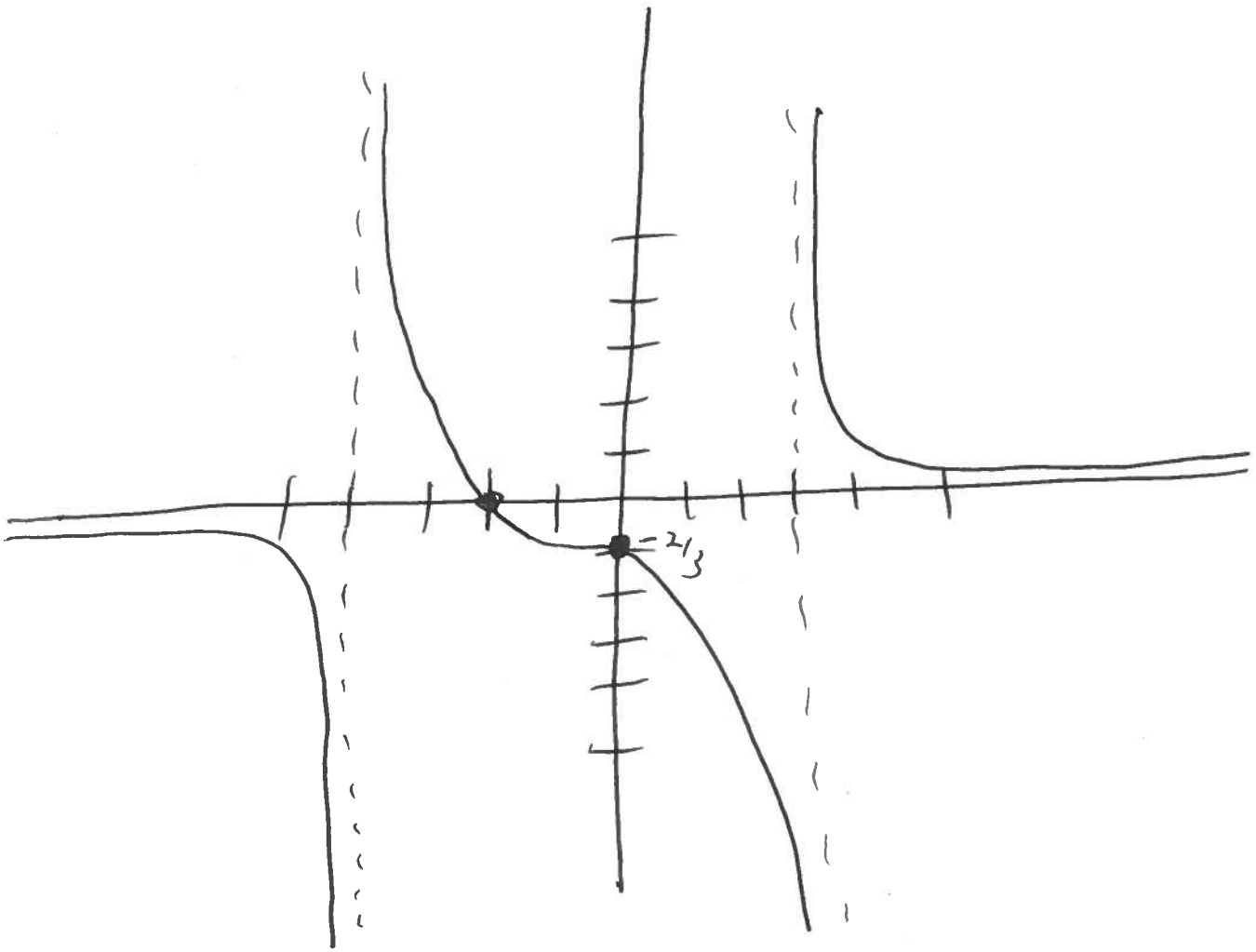
$$\Rightarrow \boxed{x=-2} \text{ x int}$$

y int

Plug in $x=0$ $\quad f(0) = \dfrac{4(0)+8}{0^2+0-12} = \dfrac{8}{-12} = \boxed{-\dfrac{2}{3}}$

For crossing the H.A. we already know since the HA is the x axis and we have the x intercept.

Sketch



Like #57 Hw #2

$$f(x) = \frac{x^2-5}{x-2}$$

V.A. where the bottom is zero

$$x-2=0 \Rightarrow x=2$$

H.A. Here the degree of the top is bigger than the degree of the bottom so we have no Horizontal assymptotes However Note that the degree of the top is only one bigger than that of the bottom. In this case we have an oblique assymptote. we find it using Polynomial Long division

$$\begin{array}{r} x+2 \\ x-2 \overline{)\, x^2+0x-5} \\ -x^2+2x \\ \hline 2x-5 \\ -2x+4 \\ \hline -1 \end{array}$$

so $f(x) = \frac{x^2-5}{x-2} = x+2+\frac{-1}{x-2}$ ←

Lets examine this

$$f(x) = \frac{x^2-5}{x-2} = x+2 - \frac{1}{x-2}$$

When x is large $x-2$ is so $\frac{1}{x-2}$ is very small close to 0

So $f(x) = \frac{x^2-5}{x-2} \approx x+2$.

$x+2$ is the equation of a line with slope 1 and y intercept 2.

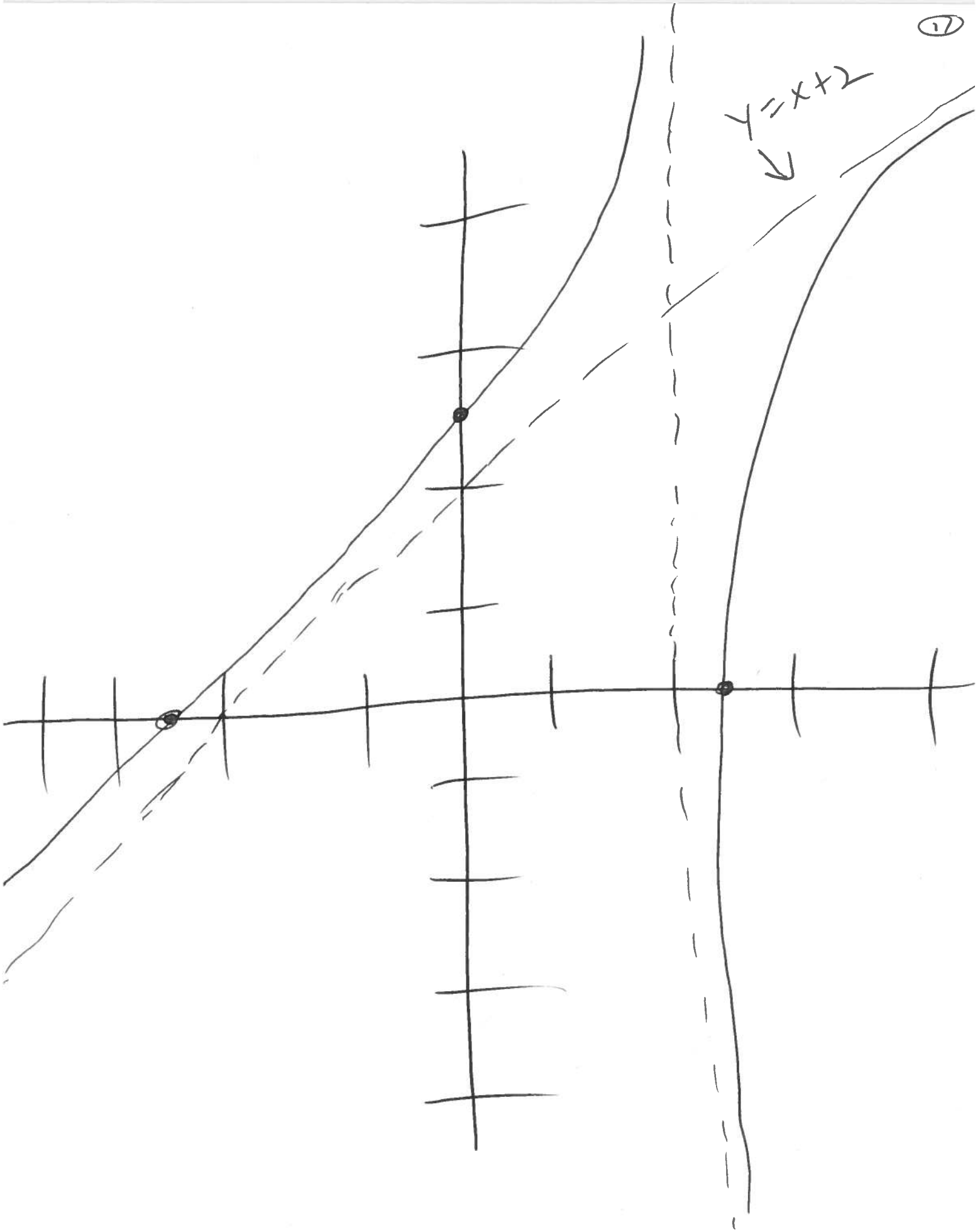
So $y=x+2$ is our oblique asymptote

x int

$$f(x)=0=\frac{x^2-5}{x-2} \Rightarrow x^2-5=0 \Rightarrow \boxed{x=\pm\sqrt{5}}$$

y int $f(0)=\frac{0^2-5}{0-2}=\frac{-5}{-2}=\boxed{5/2}$

Lets sketch


y=x+2

When $N(x)$, $D(x)$ have a common factor

Ex

Sketch $$f(x) = \frac{(x+2)(x-3)(x-2)(x+3)}{(x-3)(x+1)}$$

Here we note that we have a factor of $x-3$ on the top and Bottom In this case we only have a V.A. when $x+1=0$ So at $x=-1$ V.A. at $x=3$ we instead have a hole in the graph.

so $f(x) = \frac{(x+3)(x+2)(x-3)(x-2)}{(x-3)(x+1)} = \frac{(x+2)(x-2)(x+3)}{(x+1)}$ $x \neq 3$

now we instead sketch

$$f(x) = \frac{(x+2)(x-2)(x+3)}{(x-1)}$$ Leaving a hole at $x=3$.

V.A. are at $x = -1$

Here the degree of the top is bigger then the bottom so we have no H.A.

x int $f(x) = 0 = \dfrac{(x+2)(x-2)(x+3)}{x+1}$

$\Rightarrow (x+2)(x-2)(x+3) = 0 \Rightarrow \boxed{x = 2, -2, -3}$

y int $f(0) = \dfrac{(0+2)(0-2)(0+3)}{0+1} = \dfrac{-12}{1} = \boxed{-12}$

To find out where the hole is plug 3 into

$f(x) = \dfrac{(3+2)(3-2)(3+3)}{3+1} = \dfrac{5(1)(6)}{4} = \dfrac{30}{4}$

$= \dfrac{15}{2} = 7.5$ So at $x = 3$ $y = 7.5$ there is a hole in our graph.

(14)

